DP Biology – C1.3

Photosynthesis

- How does photosynthesis capture and use the energy in sunlight?
- How do non-living factors affect photosynthesis?

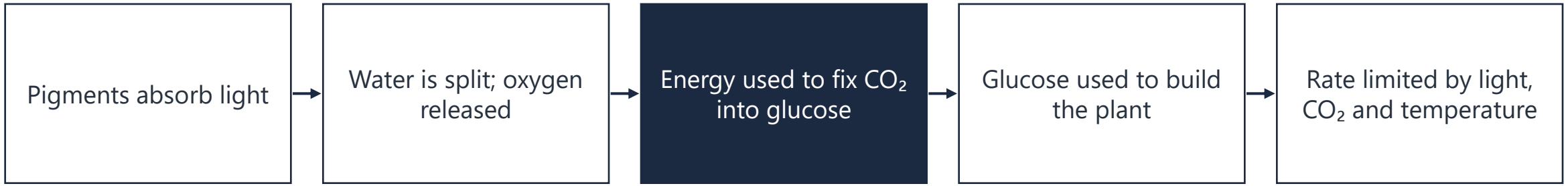
SL statements C1.3.1–8. Textbook pp. 425–438.
Image: light passing through a leaf. Jon Sullivan, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	C1.3.1–3	Explain the energy change in photosynthesis, write its word equation, and explain where the oxygen comes from
2	C1.3.4–5	Separate leaf pigments by chromatography, calculate Rf values, and explain how pigments absorb light
3	C1.3.6	Compare absorption and action spectra, using evidence such as Engelmann's experiment
4	C1.3.7–8	Investigate limiting factors, interpret data, and evaluate CO_2 enrichment experiments

The big picture: capturing sunlight



Almost all the energy that flows through living things started as sunlight captured by **photosynthesis**. Plants, algae and cyanobacteria use light energy to build carbon compounds from carbon dioxide and water.

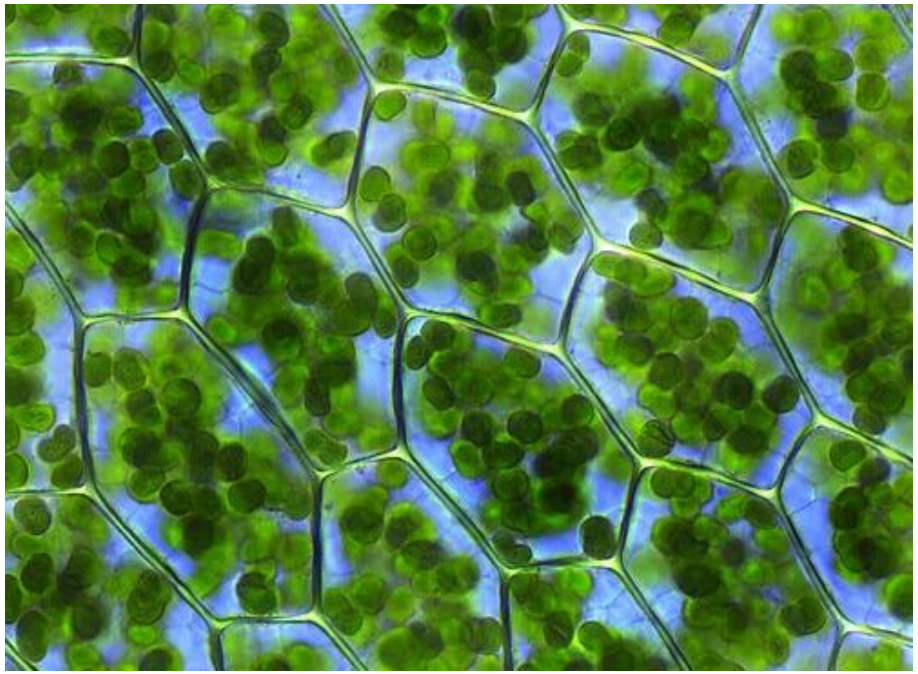
Those compounds feed the plant itself and, through food chains, almost every other organism. As a by-product, photosynthesis also produced the oxygen in the air we breathe.

Big question for this chapter

A large oak tree has a mass of many tonnes. Where did all that mass come from?

From light energy to chemical energy

C1.3.1



Chloroplasts in moss cells. Kristian Peters, CC BY-SA 3.0.

Photosynthesis transforms **light energy** into **chemical energy** stored in carbon compounds such as glucose.

In plants and algae it takes place in **chloroplasts**, the green organelles you can see in these moss cells. Cyanobacteria do it without chloroplasts.

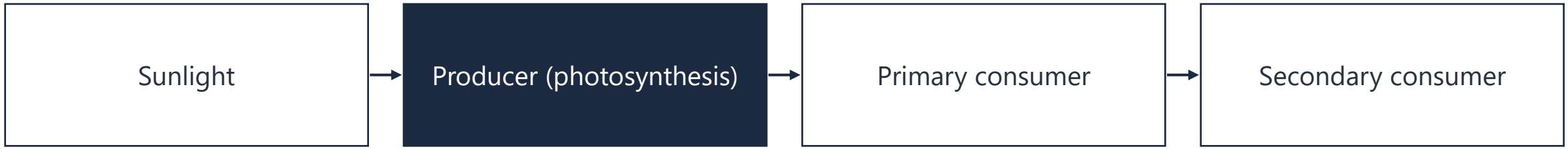
Organisms that make their own carbon compounds like this are **autotrophs**.

Key idea

Energy is not made in photosynthesis: it is transformed from one form (light) into another (chemical energy in bonds).

Photosynthesis powers ecosystems

C1.3.1



The carbon compounds made by producers are eaten by **heterotrophs**: consumers and decomposers. So nearly every food chain starts with photosynthesis.

At each step some energy is lost as heat, so ecosystems need a constant new supply of energy from sunlight. Photosynthesis is the way in.

Link

C4.2 Transfers of energy and matter: food chains, energy pyramids and the carbon cycle.

The word equation

C1.3.2

carbon dioxide + water → glucose + oxygen

(using light energy, absorbed by chlorophyll)

Light energy is used to **split water**, releasing oxygen. The hydrogen from water is used to **reduce carbon dioxide**, making glucose, which stores the energy.

Glucose can then be turned into starch for storage, sucrose for transport, cellulose for cell walls, and with mineral nutrients into amino acids and other compounds.

Big question

The mass of a tree comes almost entirely from carbon dioxide in the air and water, not from the soil.

Oxygen is a by-product

C1.3.3



Stromatolites: living mats of cyanobacteria. Happy Little Nomad, CC BY-SA 2.0.

The oxygen released comes from the **splitting of water** (photolysis).

The organism uses some of it in respiration; the rest diffuses out as a **by-product**.

Cyanobacteria were the first organisms to do this. Over hundreds of millions of years their oxygen built up in the oceans and atmosphere, changing the planet and making aerobic life possible.

Evidence that the oxygen comes from water

C1.3.3

In 1941, Ruben and Kamen gave algae water containing a heavy isotope of oxygen (oxygen-18). The oxygen gas the algae released contained oxygen-18.

When they gave the algae carbon dioxide containing oxygen-18 instead, the oxygen released did not contain it. So the oxygen released must come from **water**, not from carbon dioxide.

Labelled with oxygen-18	Oxygen gas released
Water	Contains oxygen-18
Carbon dioxide	Does not contain oxygen-18

Nature of science

Isotopes let scientists trace atoms through a reaction, an example of how new techniques lead to new understanding.

Check-in: the process of photosynthesis

C1.3.1–3

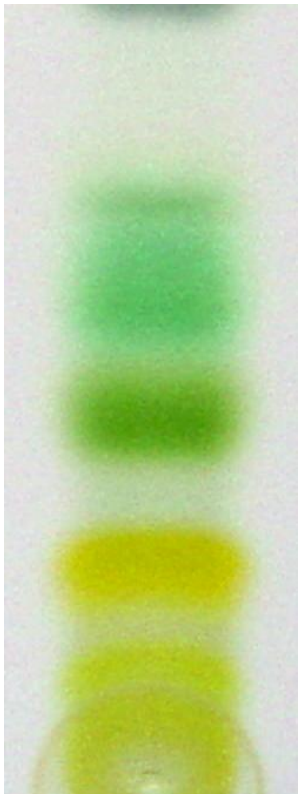
1. State the energy change that takes place in photosynthesis.
2. Write the word equation for photosynthesis.
3. State the source of the oxygen released in photosynthesis.
4. Describe the evidence that the oxygen comes from water.
5. Explain why photosynthesis matters to animals that do not photosynthesize.

How to use this

Answer without your notes, then compare with a partner.

Separating leaf pigments by chromatography

C1.3.4



Pigments on a strip. Flo, CC BY-SA 2.5.

1. Grind a leaf with a little solvent (such as propanone) to extract the pigments.
2. Put a small, concentrated spot of extract on a pencil line near the bottom of a strip of paper or a TLC plate.
3. Stand the strip in a little solvent, keeping the spot above the solvent.
4. The solvent moves up, carrying each pigment a different distance because the pigments differ in solubility.
5. Mark the solvent front before it reaches the top.

Pigment	Colour
Carotene	Orange-yellow
Xanthophyll	Yellow
Chlorophyll a	Blue-green
Chlorophyll b	Yellow-green

Safety

Use solvents in a ventilated room, away from flames, and wear eye protection.

Why pencil?

Ink would dissolve in the solvent and separate too, confusing the results.

Identifying pigments with Rf values

C1.3.4

Formula

Rf = distance moved by the pigment / distance moved by the solvent

Measure each distance from the pencil line: to the centre of the spot for the pigment, and to the solvent front for the solvent. Rf is always between 0 and 1 and has **no units**.

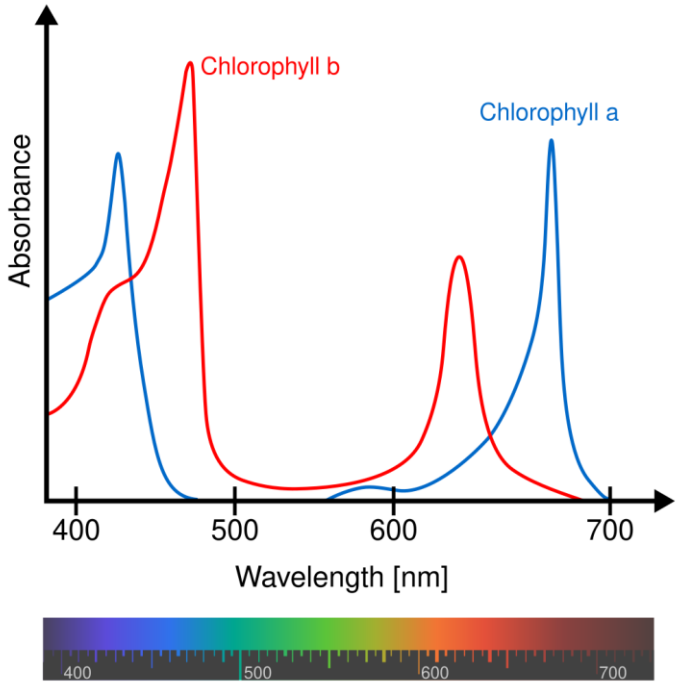
Rf depends on the solvent and the paper used, so identify each pigment by its **colour** as well as its Rf.

Worked example

The solvent front moved 80 mm. A yellow-green spot moved 36 mm. Rf = 36 / 80 = 0.45. A blue-green spot moved 48 mm: Rf = 48 / 80 = 0.60. Comparing with published values for the same solvent identifies them as chlorophyll b and chlorophyll a.

How pigments absorb light

C1.3.5



Light travels as packets of energy called **photons**. Each wavelength (colour) has a different amount of energy.

When a pigment molecule absorbs a photon, one of its **electrons is excited** to a higher energy level. Light energy has become chemical energy.

Only photons with the right energy are absorbed. **Chlorophyll** absorbs **red** and **blue** light strongly and **reflects green**, which is why leaves look green.

Absorption spectra of chlorophyll a and b. Daniele Pugliesi, CC BY-SA 3.0.

Exam tip

Label the horizontal axis of a spectrum with wavelengths (about 400 to 700 nm) and the colours.

Why leaves have several pigments

C1.3.5

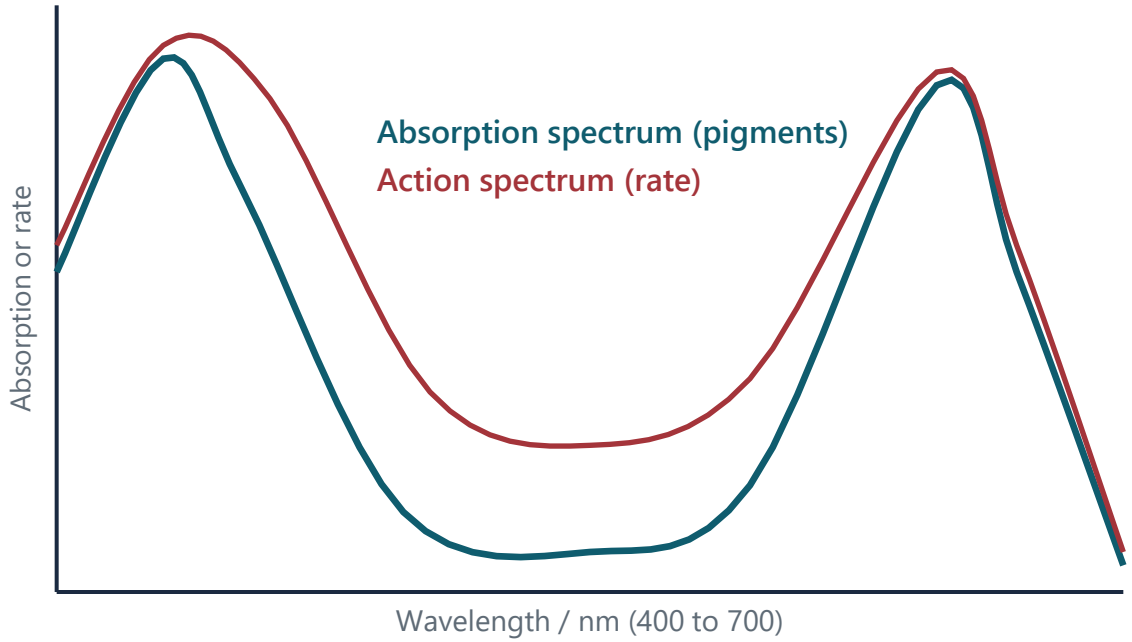
Pigment	Absorbs mainly	Role
Chlorophyll a	Blue and red	Main pigment; passes excited electrons on
Chlorophyll b	Blue and orange-red	Accessory: widens the range of light used
Carotenoids (carotene, xanthophyll)	Blue and blue-green	Accessory pigments; also protect against too much light

Autumn colours

In autumn, deciduous trees break down their chlorophyll and recycle its nutrients. The yellow and orange carotenoids, present all along, become visible. (Red colours come from new pigments made in autumn.)

Absorption and action spectra

C1.3.6



General shapes only. Blue at the left, red at the right.

	Absorption	Action
Vertical axis	% light absorbed	Rate of photosynthesis
Peaks	Blue and red	Blue and red
Green light	Very little absorbed	Low rate, not zero

Why are they similar?

Light must be absorbed to drive photosynthesis, so the action spectrum follows the absorption spectrum. It is higher in places because accessory pigments also absorb, and light scattered inside the leaf gets another chance to be absorbed.

Engelmann’s experiment: an early action spectrum

C1.3.6

In 1882, Theodor Engelmann shone a spectrum of light along a strand of green alga under a microscope, in water containing **aerobic bacteria**, which move towards oxygen.

The bacteria gathered mostly along the parts of the alga lit by **blue** and **red** light, and very few gathered in green light.

Because the bacteria follow oxygen, their numbers showed where photosynthesis was fastest: an action spectrum made with living bacteria as the detector.

Discuss

What is the independent variable and what is the dependent variable in Engelmann’s experiment? What assumption did he make about the bacteria? Why was this a clever method in 1882, before oxygen sensors existed?

Measuring the rate of photosynthesis

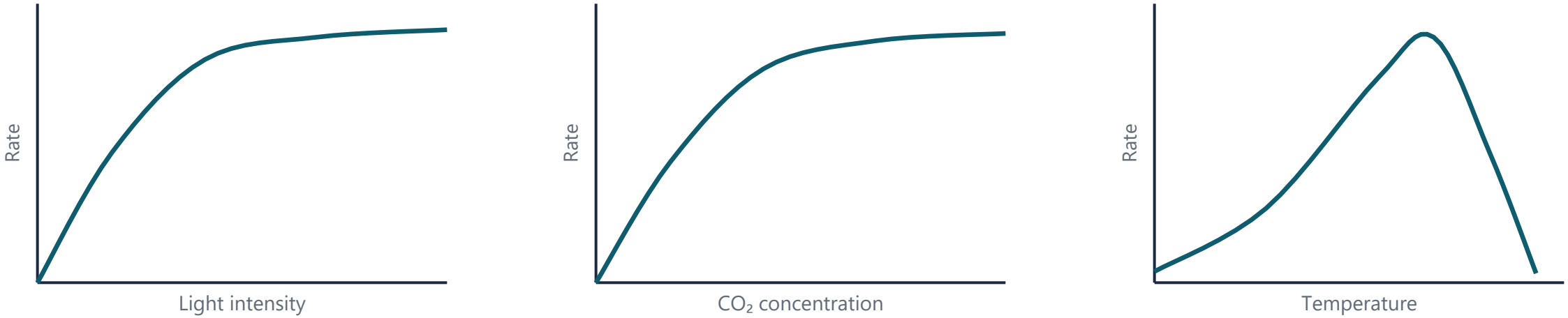
C1.3.7

Measure	How
Oxygen released	Count bubbles from pondweed, or collect the gas in a tube over a fixed time
Carbon dioxide taken in	Follow the pH of the water with an indicator: it rises as CO_2 is used
Biomass gained	Find the dry mass of plants before and after a period of growth

Factor to vary	How to change it
Light intensity	Move a lamp to measured distances (use an LED to avoid heating)
CO_2 concentration	Add different amounts of sodium hydrogencarbonate to the water
Temperature	Stand the tube of pondweed in water baths

Limiting factors

C1.3.7



General shapes only, with no numerical values.

The rate of photosynthesis is limited by whichever factor is in shortest supply: the **limiting factor**. Increasing it raises the rate, until another factor becomes limiting and the graph levels off.

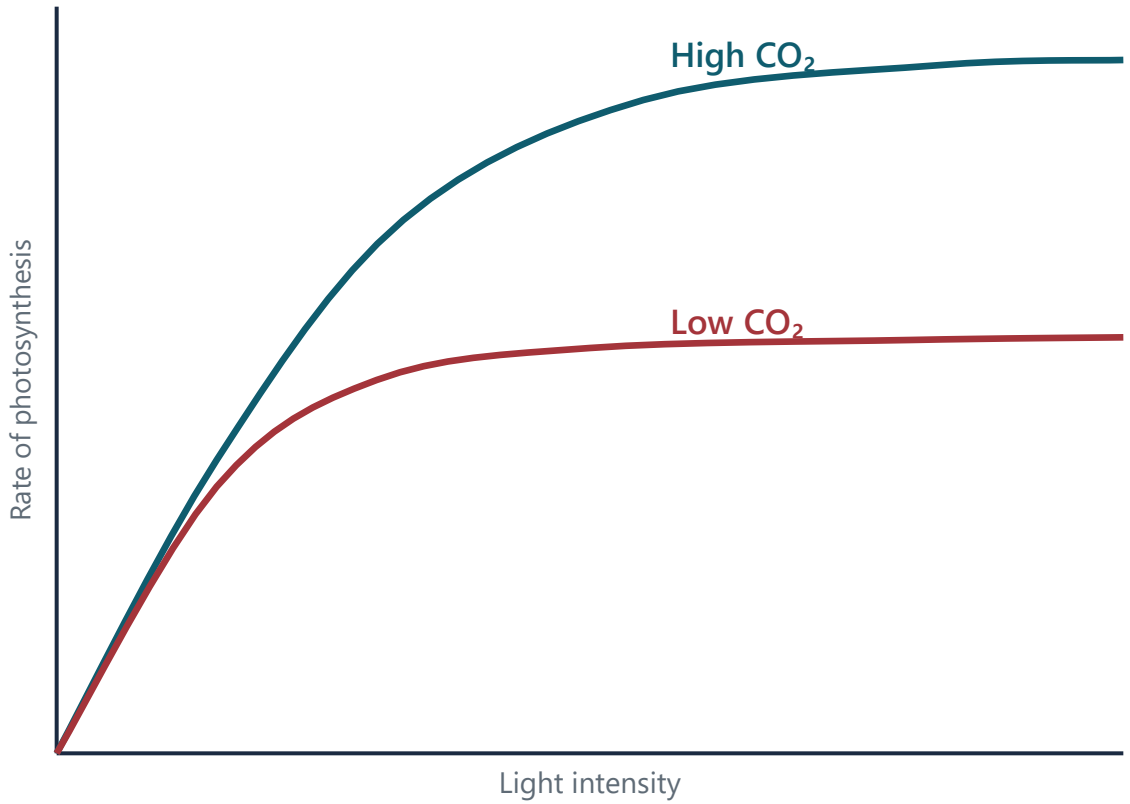
Temperature is different: photosynthesis is controlled by enzymes, so above an optimum the rate falls as they denature.

Try it

On the light-intensity graph, which factor limits the rate on the rising part? Which factors could limit it on the plateau?

Reading a graph with two limiting factors

C1.3.7



At **low light**, both lines rise together: light is limiting, so extra CO_2 makes no difference.

At **high light**, the lines separate. The low-CO_2 plant levels off first, because CO_2 has become limiting.

Raising CO_2 lifts the plateau, until light or temperature limits the rate again.

Working with data: pondweed and light

C1.3.7

Bubbles per minute from pondweed at different distances from an LED lamp, at 25 °C. These data are invented for practice; the pattern is realistic.

Distance / cm	Relative light intensity	Bubbles per minute
10	100	48
20	25	30
30	11	17
40	6	9
50	4	6

1. Describe the effect of distance on the rate.
2. Light intensity falls with the square of distance. Show how the value at 20 cm was calculated.
3. Suggest why the rate does not rise four times between 20 cm and 10 cm.
4. Suggest one improvement to the method.

Carbon dioxide enrichment experiments

C1.3.8



A greenhouse. Lolo425, CC BY-SA 3.0.



A FACE facility in Nevada. NNSA, public domain.

CO_2 in the atmosphere is rising. Will plants grow faster and absorb more of it? Enrichment experiments raise CO_2 and measure growth.

	Enclosed greenhouse	FACE (free-air CO_2 enrichment)
Method	Add CO_2 to a closed space	Release CO_2 over open plots in the field
Strength	Other factors easy to control	Real conditions: many species, real soil and weather

Most studies find that growth rises at first, but by less than expected, and often levels off as other factors (such as nitrogen or water) become limiting.

Check-in: pigments, spectra and limiting factors

C1.3.4–8

1. A pigment moves 24 mm and the solvent front 60 mm. Calculate the Rf value.
2. Explain why chlorophyll looks green.
3. Distinguish between an absorption spectrum and an action spectrum.
4. Explain the term limiting factor, using light intensity as an example.
5. Outline one advantage of FACE experiments over greenhouse experiments.

How to use this

Write full sentences, then mark with a partner.

Common mistakes to avoid

Not this	But this
Plants get their mass from the soil.	Most of a plant's mass comes from carbon dioxide (and water).
The oxygen released comes from carbon dioxide.	It comes from the splitting of water.
Chlorophyll absorbs green light.	It reflects green light and absorbs red and blue.
Plants photosynthesize instead of respiring.	Plants do both; photosynthesis only in light, respiration all the time.
Rf has units of mm.	Rf is a ratio, so it has no units.

Check that you can...

1. **State** the energy change in photosynthesis and its importance to ecosystems
2. **Write** the word equation for photosynthesis
3. **Explain** why oxygen is a by-product, and the evidence that it comes from water
4. **Describe** how to separate pigments by chromatography, and **calculate** Rf values
5. **Explain** how pigments absorb light, and why plants have several pigments
6. **Compare** absorption and action spectra
7. **Explain** limiting factors and **interpret** graphs and data
8. **Evaluate** CO_2 enrichment experiments

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Write the word equation for photosynthesis.

2. A pigment moves 24 mm, and the solvent moves 60 mm. Calculate Rf.

3. State where the oxygen released in photosynthesis comes from.

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